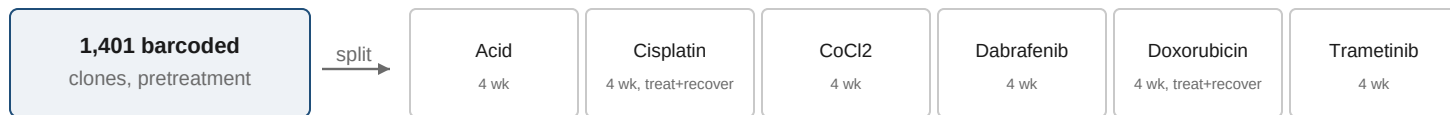


Figure 1 Clone-level prospective design and evaluable population

One BRAF-V600E melanoma line (WM989). Data from Schaff et al. 2026; reanalysis only.

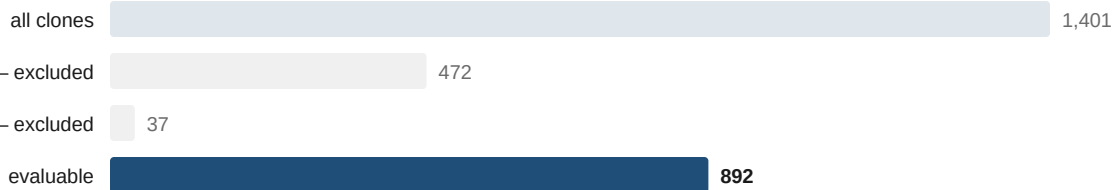


Outcome C1 post-treatment clone DETECTION — an observed zero means no assigned cell was seen.

Not death, not sensitivity, not clinical response.

Evaluable population

A within-clone AUROC is undefined without both a detected and an undetected condition.



Closed vocabulary

56 of 56 adversarial condition strings refused, including 16 oncology drugs (Vemurafenib, Carboplatin, ...).

Design width 309 columns = 50 PCs + 4 abundance + 5 condition + 250 interaction.